Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Jelle Van Der Hoek

Physical address of dairy:

Number and Street	City	County	Zip Code
337 Gates RD	Modesto	Stanislaus	95358

Street and nearest cross street (if no address): Gates & Hw. 132

Date facility was originally placed in operation: 01/15/1945

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X012-X044-X012-XXXX

B. OPERATORS

Van Der Hoek, Jelle

Operator name: Van Der Hoek, Jelle

Telephone no.: (209) 523-9343 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
337 Gates RD	Modesto	CA	95358

Van Der Hoek, Michael

Operator name: Van Der Hoek, Michael

Telephone no.: (209) 523-9343 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
337 Gates RD	Modesto	CA	95358

This operator is responsible for paying permit fees.

Van Der Hoek Dairies

Operator name: Van Der Hoek Dairies

Telephone no.: (209) 523-9343 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
337 Gates RD	Modesto	CA	95358

C. OWNERS

Van Der Hoek, Jelle

Legal owner name:	Van Der Hoek, Jelle	Telephone no.:	(209) 523-9343	
			Landline	Cellular
337 Gates RD	Modesto		CA	95358
Mailing Address Number and Street	City		State	Zip Code

Van Der Hoek, Michael

Legal owner name:	Van Der Hoek, Michael	Telephone no.:	(209) 523-9343	
			Landline	Cellular
337 Gates RD	Modesto		CA	95358
Mailing Address Number and Street	City		State	Zip Code

This owner is responsible for paying permit fees.

Van Der Hoek Dairies

Legal owner name:	Van Der Hoek Dairies	Telephone no.:	(209) 523-9343	
			Landline	Cellular
337 Gates RD	Modesto		CA	95358
Mailing Address Number and Street	City		State	Zip Code

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	90
Number under roof	650	50	270	210	90	0
Maximum number	650	50	270	210	90	90
Average number	650	50	270	210	90	90
Avg live weight (lbs)	1,000	1,100	800	450		

Predominant milk cow breed: Jersey

Average milk production: 65 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 22,035.04 *tons per reporting period*

Total nitrogen from manure: 276,885.07 *lbs per reporting period*

After ammonia losses (30% loss applied): 193,819.55 *lbs per reporting period*

Total phosphorus from manure: 45,552.35 *lbs per reporting period*

Total potassium from manure: 122,345.75 *lbs per reporting period*

Total salt from manure: 317,550.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 4,104,000 *gallons*

Total nitrogen generated: 27,572.55 *lbs*

Total phosphorus generated: 2,586.74 *lbs*

Total potassium generated: 48,427.70 *lbs*

Total salt generated: 289,508.75 *lbs*

	4,104,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	4,104,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
AG Well	Ground water
IW-North	Ground water
MID Lateral 3	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
04/15/2023	Solid commercial fertilizer *UN 32*	8.21 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	5,249.15	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	5,249.15	0.00	0.00	0.00

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
01/26/2023	Corral solids	51.90 *ton*	Dry-weight	8.8		19,100.00	4,350.00	18,400.00		43.80
03/28/2023	Corral solids	46.40 *ton*	Dry-weight	8.8		19,100.00	4,350.00	18,400.00		43.80
05/19/2023	Corral solids	600.00 *ton*	Dry-weight	8.8		19,100.00	4,350.00	18,400.00		43.80
09/18/2023	Corral solids	70.00 *ton*	Dry-weight	48.0		19,300.00	4,360.00	7,920.00		36.80

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	25,732.69	5,858.00	24,012.64	584,670.65
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	25,732.69	5,858.00	24,012.64	584,670.65

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Dairy Almonds	5	5	1	manure	X012-X044-X013-XXXX
Field 1	49	49	2	process wastewater	X012-X044-X012-XXXX X012-X044-X013-XXXX
Newman Almonds	85	85	1	manure	X026-X014-X009-XXXX X026-X014-X038-XXXX
Newman Walnuts	41	41	1	none	X026-X014-X039-XXXX
West	51	51	1	none	X012-X044-X005-XXXX
Totals for areas that were used for application	139	139	4		
Totals for areas that were not used for application	92	92	2		
Land application area totals	231	231	6		

B. CROPS AND HARVESTS

Dairy Almonds

Field name: Dairy Almonds

10/23/2017: Almonds Field Run

Crop: Almonds Field Run Acres planted: 5 Plant date: 10/23/2017

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	34.00 *ton*	As-is		11.6	15,900.00	2,260.00	16,000.00		7.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	259.80	43.80	282.00	0.00
Total actual harvest content	6.80	216.24	30.74	217.60	889.66

Field 1

Field name: Field 1

Field 1

11/12/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 49 Plant date: 11/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	882.00 *ton*	As-is		68.8	3,590.00	1,200.00	5,570.00		8.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.00	129.24	43.20	200.52	965.95

06/22/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 49 Plant date: 06/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/17/2023	951.00 *ton*	As-is		63.4	7,140.00	820.00	15,500.00		16.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	216.00	0.00
Total actual harvest content	19.41	277.15	31.83	601.65	2,344.12

Newman Almonds

Field name: Newman Almonds

01/01/2008: Almonds Field Run

Crop: Almonds Field Run Acres planted: 85 Plant date: 01/01/2008

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/17/2023	603.00 *ton*	As-is		11.6	15,900.00	2,260.00	16,000.00		7.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	259.80	43.80	282.00	0.00
Total actual harvest content	7.09	225.59	32.07	227.01	928.14

Newman Walnuts

Field name: Newman Walnuts

01/01/1996: Walnuts

Crop: Walnuts Acres planted: 41 Plant date: 01/01/1996

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/22/2023	96.00 *ton*	As-is		10.0	20,000.00	1,500.00	14,400.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	2.50	100.00	7.50	72.00	0.00
Total actual harvest content	2.34	93.66	7.02	67.43	0.00

West

Field name: West

02/08/2017: Almonds Field Run

Crop: Almonds Field Run Acres planted: 51 Plant date: 02/08/2017

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/21/2023	388.00 *ton*	As-is		11.6	15,900.00	2,260.00	16,000.00		7.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	259.80	43.80	282.00	0.00
Total actual harvest content	7.61	241.93	34.39	243.45	995.35

NUTRIENT BUDGET

A. LAND APPLICATIONS

Dairy Almonds - 10/23/2017: Almonds Field Run

Field name: Dairy Almonds

Crop: Almonds Field Run

Plant date: 10/23/2017

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/16/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Compost	Compost	133.37	41.29	267.32	1,619.07	20.00 *ton*
Application event totals		133.37	41.29	267.32	1,619.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG Well	Ground water	9.19	0.00	0.00	298.41	483,236.00 *gal*
Application event totals		9.19	0.00	0.00	298.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/22/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	15.36	0.00	0.00	0.00	
Application event totals		15.36	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG Well	Ground water	18.39	0.00	0.00	596.83	966,472.00 *gal*
Application event totals		18.39	0.00	0.00	596.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG Well	Ground water	18.39	0.00	0.00	596.83	966,472.00 *gal*
Application event totals		18.39	0.00	0.00	596.83	

Dairy Almonds - 10/23/2017: Almonds Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	15.36	0.00	0.00	0.00	
Application event totals		15.36	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG Well	Ground water	18.39	0.00	0.00	596.83	966,472.00 *gal*
Application event totals		18.39	0.00	0.00	596.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG Well	Ground water	9.19	0.00	0.00	298.41	483,236.00 *gal*
Application event totals		9.19	0.00	0.00	298.41	

Field 1 - 11/12/2022: Oats, silage-soft dough

Field name: Field 1

Crop: Oats, silage-soft dough

Plant date: 11/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	80.12	9.44	175.35	884.83	792,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	0.00	5,940,000.00 *gal*
Application event totals		80.12	9.44	175.35	884.83	

Field 1 - 11/12/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	97.48	13.49	137.33	686.67	720,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	34.03	5,400,000.00 *gal*
Application event totals		97.48	13.49	137.33	720.70	

Field 1 - 06/22/2023: Sudangrass, silage

Field name: Field 1

Crop: Sudangrass, silage

Plant date: 06/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	181.97	16.48	219.74	1,342.45	1,008,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	47.64	7,560,000.00 *gal*
Application event totals		181.97	16.48	219.74	1,390.09	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	40.83	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.83	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	101.57	6.69	227.95	1,497.20	792,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	37.43	5,940,000.00 *gal*
Application event totals		101.57	6.69	227.95	1,534.63	

Field 1 - 06/22/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	101.57	6.69	227.95	1,497.20	792,000.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	37.43	5,940,000.00 *gal*
Application event totals		101.57	6.69	227.95	1,534.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	37.43	5,940,000.00 *gal*
Application event totals		0.00	0.00	0.00	37.43	

Newman Almonds - 01/01/2008: Almonds Field Run

Field name: Newman Almonds

Crop: Almonds Field Run

Plant date: 01/01/2008

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/18/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Compost	Compost	137.29	42.51	275.18	1,666.69	350.00 *ton*
Application event totals		137.29	42.51	275.18	1,666.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG Well	Ground water	9.23	0.00	0.00	299.55	8,246,269.00 *gal*
Application event totals		9.23	0.00	0.00	299.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	15.36	0.00	0.00	0.00	
Application event totals		15.36	0.00	0.00	0.00	

Newman Almonds - 01/01/2008: Almonds Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG Well	Ground water	18.46	0.00	0.00	599.10	16,492,538.00 *gal*
Application event totals		18.46	0.00	0.00	599.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG Well	Ground water	18.46	0.00	0.00	599.10	16,492,538.00 *gal*
Application event totals		18.46	0.00	0.00	599.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	15.36	0.00	0.00	0.00	
Application event totals		15.36	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG Well	Ground water	18.46	0.00	0.00	599.10	16,492,538.00 *gal*
Application event totals		18.46	0.00	0.00	599.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG Well	Ground water	9.23	0.00	0.00	299.55	8,246,269.00 *gal*
Application event totals		9.23	0.00	0.00	299.55	

Newman Walnuts - 01/01/1996: Walnuts

Field name: Newman Walnuts

Crop: Walnuts

Plant date: 01/01/1996

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Newman Walnuts - 01/01/1996: Walnuts

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG Well	Ground water	9.88	0.00	0.00	320.52	4,256,129.00 *gal*
Application event totals		9.88	0.00	0.00	320.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	17.60	0.00	0.00	0.00	
Application event totals		17.60	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG Well	Ground water	19.75	0.00	0.00	641.05	8,512,258.00 *gal*
Application event totals		19.75	0.00	0.00	641.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG Well	Ground water	19.75	0.00	0.00	641.05	8,512,258.00 *gal*
Application event totals		19.75	0.00	0.00	641.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	17.60	0.00	0.00	0.00	
Application event totals		17.60	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG Well	Ground water	19.75	0.00	0.00	641.05	8,512,258.00 *gal*
Application event totals		19.75	0.00	0.00	641.05	

Newman Walnuts - 01/01/1996: Walnuts

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG Well	Ground water	9.88	0.00	0.00	320.52	4,256,129.00 *gal*
Application event totals		9.88	0.00	0.00	320.52	

West - 02/08/2017: Almonds Field Run

Field name: West

Crop: Almonds Field Run

Plant date: 02/08/2017

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-North	Ground water	37.90	0.00	0.00	737.51	4,462,629.00 *gal*
Application event totals		37.90	0.00	0.00	737.51	

04/22/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	10.24	0.00	0.00	0.00	
Application event totals		10.24	0.00	0.00	0.00	

05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-North	Ground water	75.80	0.00	0.00	1,475.02	8,925,258.00 *gal*
Application event totals		75.80	0.00	0.00	1,475.02	

06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-North	Ground water	75.80	0.00	0.00	1,475.02	8,925,258.00 *gal*
Application event totals		75.80	0.00	0.00	1,475.02	

West - 02/08/2017: Almonds Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	10.24	0.00	0.00	0.00	
Application event totals		10.24	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-North	Ground water	75.80	0.00	0.00	1,475.02	8,925,258.00 *gal*
Application event totals		75.80	0.00	0.00	1,475.02	

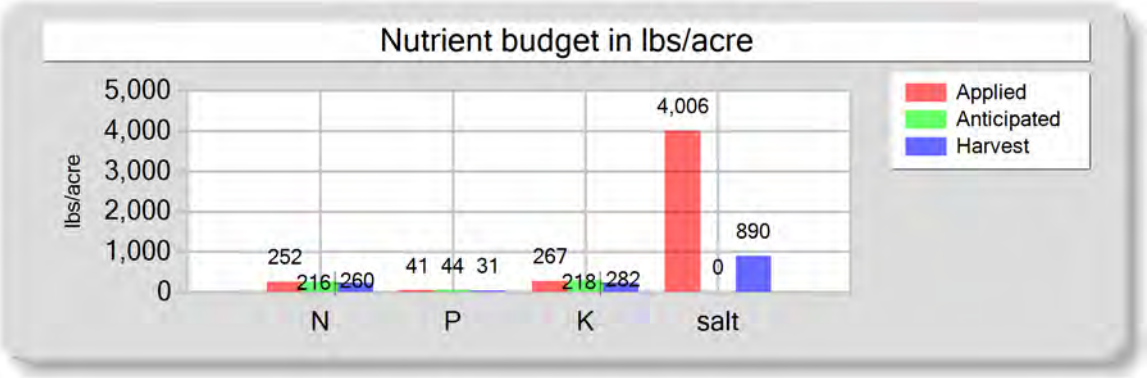
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-North	Ground water	37.90	0.00	0.00	737.51	4,462,629.00 *gal*
Application event totals		37.90	0.00	0.00	737.51	

B. NUTRIENT BUDGET

Dairy Almonds - 10/23/2017: Almonds Field Run

Field name: Dairy Almonds Crop: Almonds Field Run Plant date: 10/23/2017



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	30.72	0.00	0.00	0.00
Dry manure	133.37	41.29	267.32	1,619.07
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	73.55	0.00	0.00	2,387.30
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	251.64	41.29	267.32	4,006.37
Anticipated crop nutrient removal	259.80	43.80	282.00	0.00
Actual crop nutrient removal	216.24	30.74	217.60	889.66
Nutrient balance	35.40	10.56	49.72	3,116.72
Applied to removed ratio	1.16	1.34	1.23	4.50

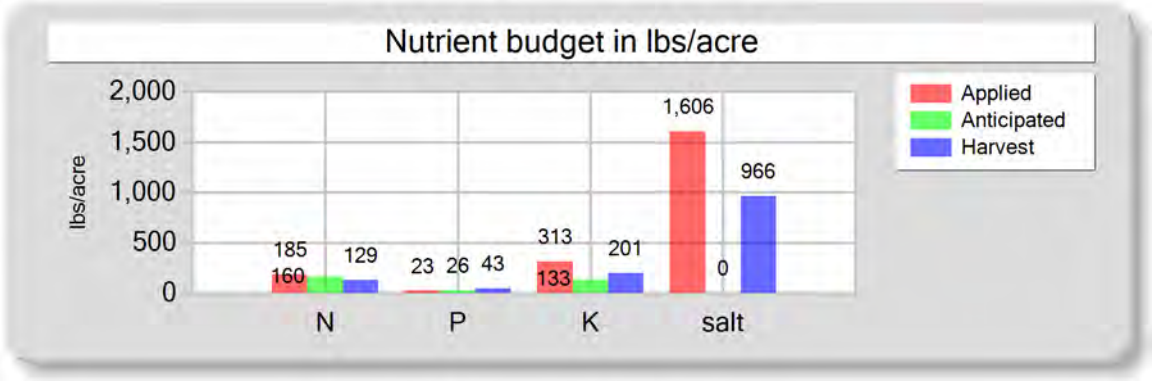
Fresh water applied	
3,865,888.00	*gallons*
142.37	*acre-inches*
28.47	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 11/12/2022: Oats, silage-soft dough

Field name: Field 1 Crop: Oats, silage-soft dough Plant date: 11/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	177.60	22.93	312.68	1,571.50
Fresh water	0.00	0.00	0.00	34.03
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	184.60	22.93	312.68	1,605.53
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	129.24	43.20	200.52	965.95
Nutrient balance	55.36	-20.27	112.16	639.58
Applied to removed ratio	1.43	0.53	1.56	1.66

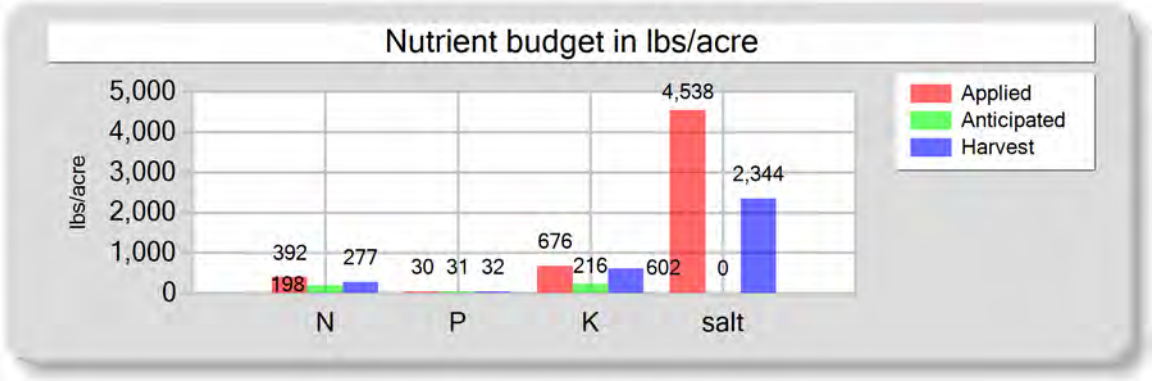
Fresh water applied	
11,340,000.00	*gallons*
417.61	*acre-inches*
8.52	*inches/acre*

Process wastewater applied	
1,512,000.00	*gallons*
55.68	*acre-inches*
1.14	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 06/22/2023: Sudangrass, silage

Field name: Field 1 Crop: Sudangrass, silage Plant date: 06/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	385.10	29.86	675.64	4,336.84
Fresh water	0.00	0.00	0.00	200.76
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	392.10	29.86	675.64	4,537.60
Anticipated crop nutrient removal	198.00	30.60	216.00	0.00
Actual crop nutrient removal	277.15	31.83	601.65	2,344.12
Nutrient balance	114.95	-1.97	73.99	2,193.48
Applied to removed ratio	1.41	0.94	1.12	1.94

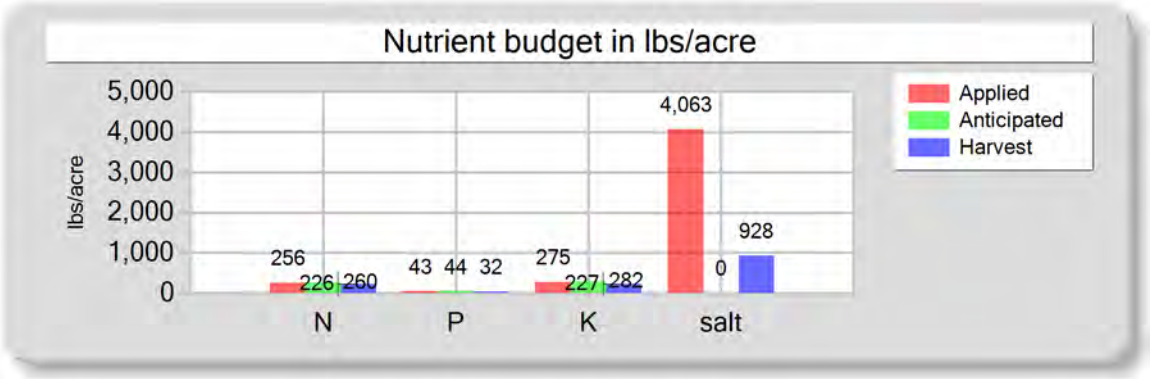
Fresh water applied	
31,860,000.00	*gallons*
1,173.30	*acre-inches*
23.94	*inches/acre*

Process wastewater applied	
2,592,000.00	*gallons*
95.45	*acre-inches*
1.95	*inches/acre*

Total harvests for the crop	
1	*harvests*

Newman Almonds - 01/01/2008: Almonds Field Run

Field name: Newman Almonds Crop: Almonds Field Run Plant date: 01/01/2008



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	30.72	0.00	0.00	0.00
Dry manure	137.29	42.51	275.18	1,666.69
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	73.83	0.00	0.00	2,396.39
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	255.85	42.51	275.18	4,063.08
Anticipated crop nutrient removal	259.80	43.80	282.00	0.00
Actual crop nutrient removal	225.59	32.07	227.01	928.14
Nutrient balance	30.25	10.44	48.17	3,134.94
Applied to removed ratio	1.13	1.33	1.21	4.38

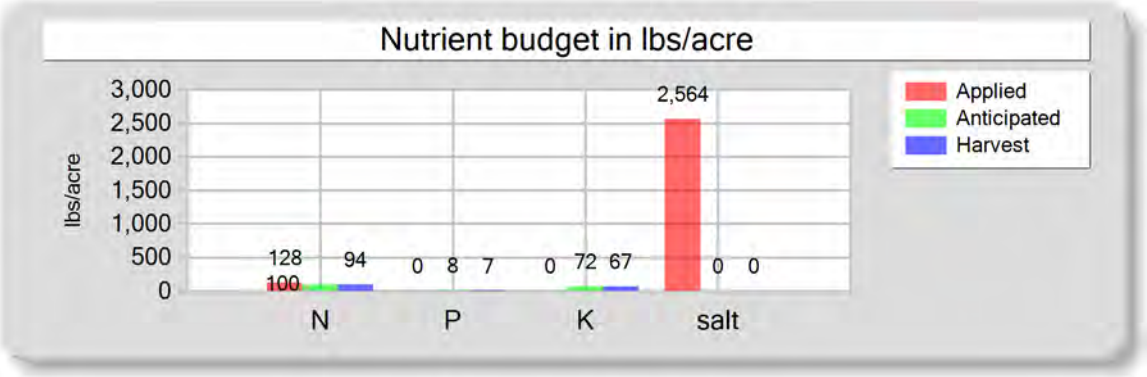
Fresh water applied	
65,970,152.00	*gallons*
2,429.46	*acre-inches*
28.58	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Newman Walnuts - 01/01/1996: Walnuts

Field name: Newman Walnuts Crop: Walnuts Plant date: 01/01/1996



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	35.20	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	79.00	0.00	0.00	2,564.18
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	128.20	0.00	0.00	2,564.18
Anticipated crop nutrient removal	100.00	7.50	72.00	0.00
Actual crop nutrient removal	93.66	7.02	67.43	0.00
Nutrient balance	34.55	-7.02	-67.43	2,564.18
Applied to removed ratio	1.37	0.00	0.00	0.00

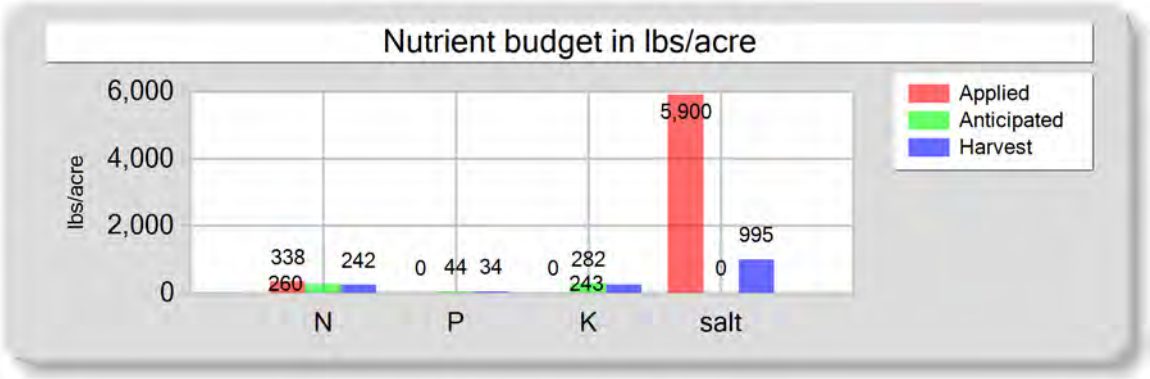
Fresh water applied	
34,049,032.00	*gallons*
1,253.91	*acre-inches*
30.58	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

West - 02/08/2017: Almonds Field Run

Field name: West | Crop: Almonds Field Run | Plant date: 02/08/2017



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	20.48	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	303.18	0.00	0.00	5,900.09
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	337.66	0.00	0.00	5,900.09
Anticipated crop nutrient removal	259.80	43.80	282.00	0.00
Actual crop nutrient removal	241.93	34.39	243.45	995.35
Nutrient balance	95.73	-34.39	-243.45	4,904.74
Applied to removed ratio	1.40	0.00	0.00	5.93

Fresh water applied	
35,701,032.00	*gallons*
1,314.75	*acre-inches*
25.78	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Compost

Sample and source description: Compost

Sample date: 10/14/2022 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 27.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,900.00	7,090.00	45,900.00							27.80
DL	100.00	100.00	100.00							0.01

Corral

Sample and source description: Corral

Sample date: 04/28/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,100.00	4,350.00	18,400.00							43.80
DL	100.00	100.00	100.00							0.01

Separator

Sample and source description: Separator

Sample date: 04/28/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,900.00	1,630.00	4,640.00							11.50
DL	100.00	100.00	100.00							0.01

Corral

Sample and source description: Corral

Sample date: 10/13/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 48.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,300.00	4,360.00	7,920.00							36.80
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

4th Qtr WW

Sample and source description: 4th Qtr WW

Sample date: 10/21/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	594.00	316.00	0.00	0.00	70.00	1,300.00								8,420.00	6,560
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

1st Qtr WW

Sample and source description: 1st Qtr WW

Sample date: 02/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	795.00	440.00	0.00	0.00	110.00	1,120.00								8,000.00	5,600
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2nd Qtr WW

Sample and source description: 2nd Qtr WW

Sample date: 05/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,060.00	591.00	0.00	0.00	96.00	1,280.00	229.00	140.00	279.00	4,960.00	0.00	289.00	239.00	9,810.00	7,820
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

3rd Qtr WW

Sample and source description: 3rd Qtr WW

Sample date: 07/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	753.00	490.00	0.00	0.00	49.60	1,690.00								9,720.00	11,100
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

4th Qtr WW

Sample and source description: 4th Qtr WW

Sample date: 10/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	664.00	468.00	0.00	0.00	63.00	1,140.00								8,990.00	5,760
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

AG Well

AG Well

AG Well

Sample description: AG Well

Sample date: 03/17/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	11.40	0.00	2.96								667.00	370
DL	0.50	0.50	0.50								1.00	10

IW-North

IW-North

Sample description: IW-North

Sample date: 08/29/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	51.90	0.00	51.60								1,680.00	1,010
DL	0.50	0.50	0.50								1.00	10

MID Lateral 3

MID Lateral 3

Sample description: MID Lateral 3

Sample date: 07/22/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	0
DL	0.50	0.50	0.50								1.00	10

MID Lateral 3

MID Lateral 3

Sample description: MID Lateral 3

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								30.70	37
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Dairy Almonds - 10/23/2017: Almonds Field Run

Almonds

Sample and source description: Almonds

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 11.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,900.00	2,260.00	16,000.00		7.40
DL	100.00	100.00	100.00		0.01

Field 1 - 11/12/2022: Oats, silage-soft dough

Field 1 - 11/12/2022: Oats, silage-soft dough

1A

Sample and source description: 1A

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,970.00	1,170.00	5,660.00		8.60
DL	100.00	100.00	100.00		0.01

1B

Sample and source description: 1B

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,590.00	1,200.00	5,570.00		8.60
DL	100.00	100.00	100.00		0.01

Field 1 - 06/22/2023: Sudangrass, silage

1A

Sample and source description: 1A

Sample date: 09/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,600.00	880.00	1,420.00		11.60
DL	100.00	100.00	100.00		0.01

Field 1 - 06/22/2023: Sudangrass, silage

1B

Sample and source description: 1B

Sample date: 09/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,140.00	820.00	15,500.00		16.50
DL	100.00	100.00	100.00		0.01

Newman Almonds - 01/01/2008: Almonds Field Run

Almonds

Sample and source description: Almonds

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 11.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,900.00	2,260.00	16,000.00		7.40
DL	100.00	100.00	100.00		0.01

West - 02/08/2017: Almonds Field Run

Almonds

Sample and source description: Almonds

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 11.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,900.00	2,260.00	16,000.00		7.40
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

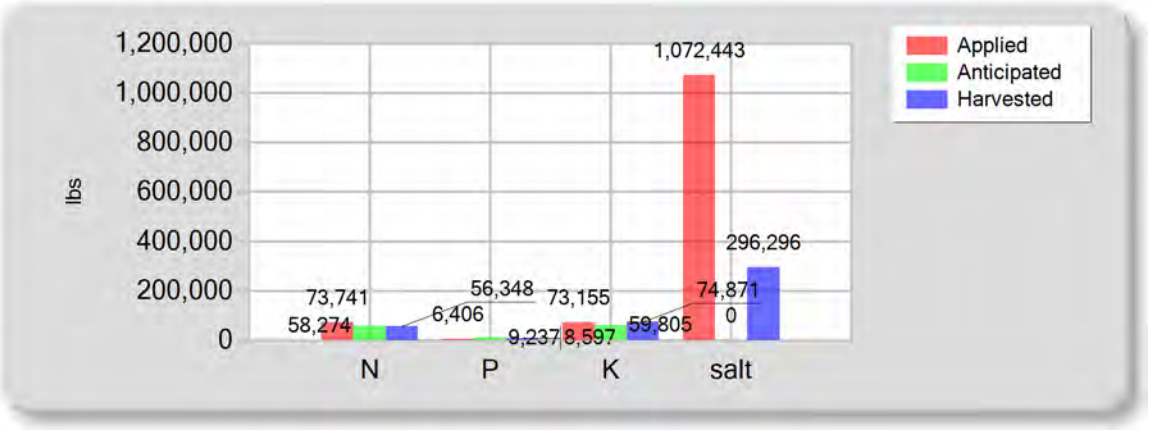
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

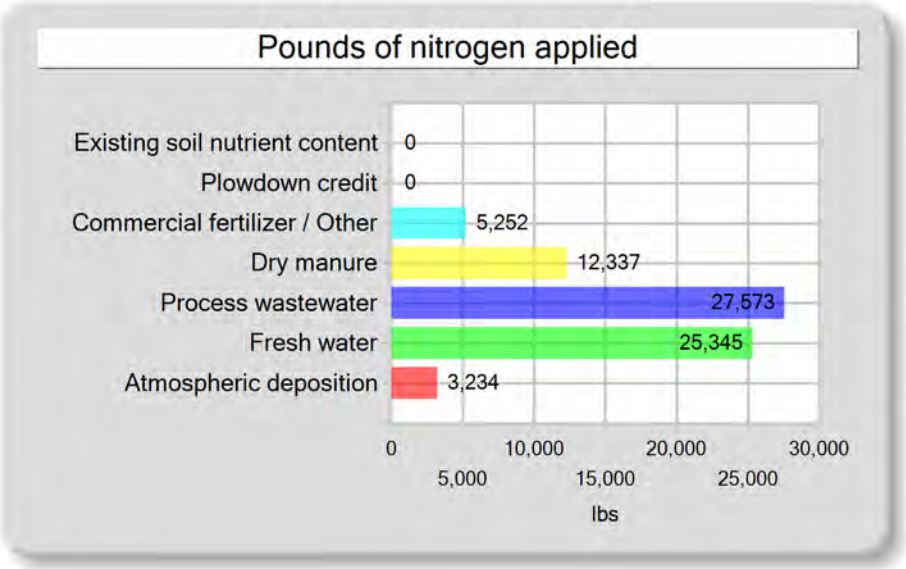
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

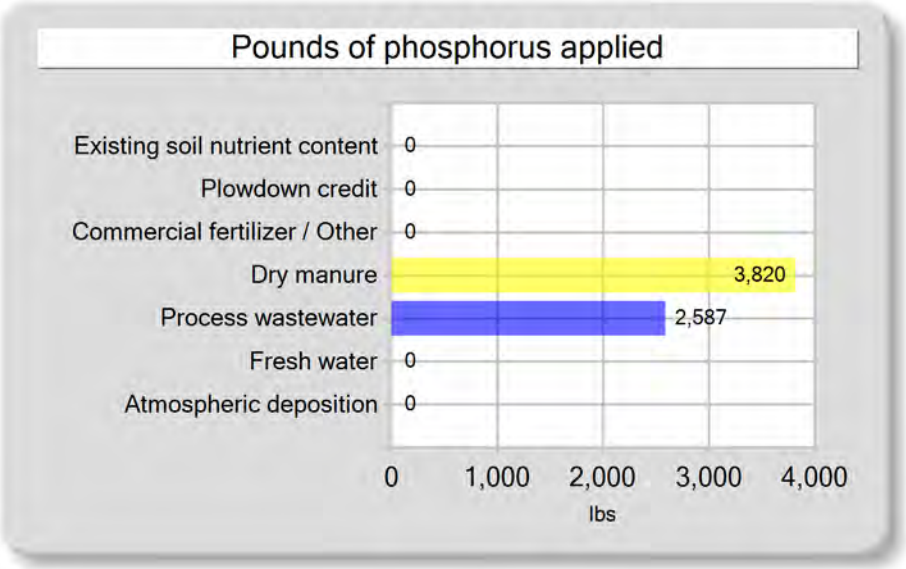
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	5,252.48	0.00	0.00	0.00
Dry manure	12,336.69	3,819.52	24,727.25	149,764.16
Process wastewater	27,572.55	2,586.74	48,427.70	289,508.75
Fresh water	25,345.21	0.00	0.00	633,169.69
Atmospheric deposition	3,234.00	0.00	0.00	0.00
Total nutrients applied	73,740.93	6,406.26	73,154.95	1,072,442.60
Anticipated crop nutrient removal	58,273.80	9,237.10	59,805.20	0.00
Actual crop nutrient removal	56,348.04	8,597.44	74,871.28	296,296.23
Nutrient balance	17,392.89	-2,191.18	-1,716.33	776,146.37
Applied to removed ratio	1.31	0.75	0.98	3.62

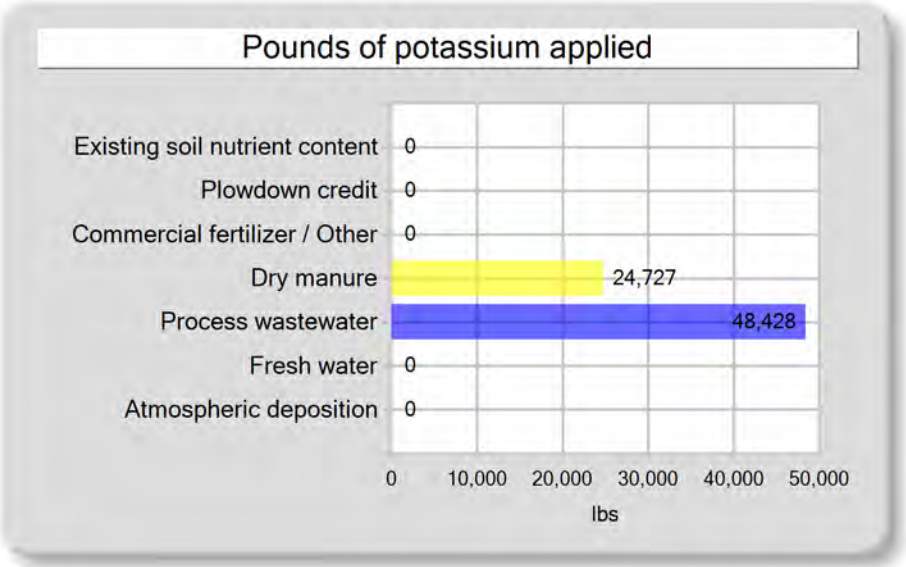
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

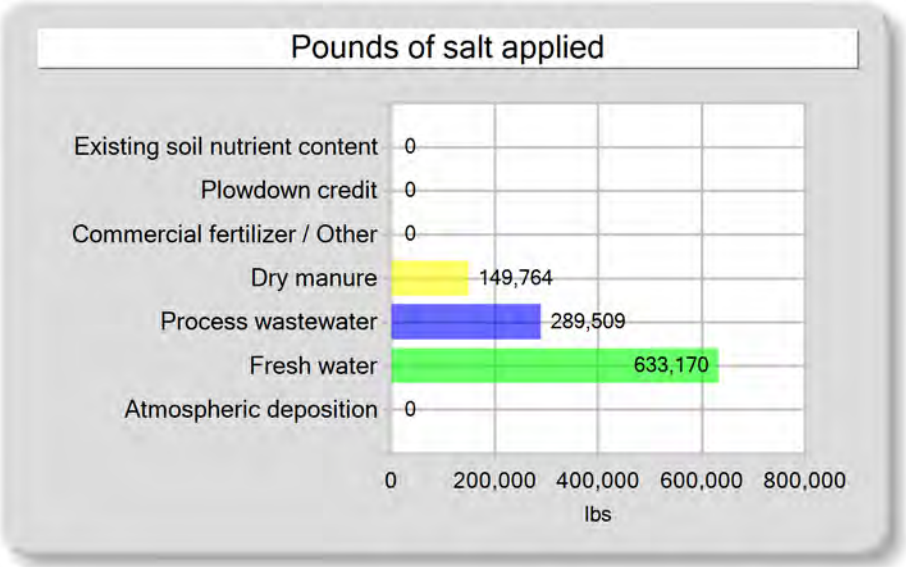


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Michael VanDerHoek (Jun 27, 2024 05:10 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Michael Van Der Hoek	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 27, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Michael Van Der Hoek

Name of Dairy Facility: Jelle Van Der Hoek

Facility Address:

337 Gates RD	Modesto	Stanislaus	95358
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number:

Michael Van Der Hoek	(209) 581-6133
Name	Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: California Soils

Address of Hauling Company/Person:

3401 Gaffery RD	Vernalis	CA	95385
Number and Street	City	State	Zip Code

Contact Person:

Office	(209) 835-9530
Name	Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Composting Facility

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

California Soils	(209) 835-9530
Name	Phone Number

3401 Gaffery RD	Vernalis	CA	95385
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

3401 Gaffery RD	Vernalis	95385
Address	City	Zip Code

	Stanislaus
Street and nearest cross street (if no address)	County

Assessor's Parcel Number	Assessor's Parcel Number County

Last date hauled: 01/26/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 51.90 tons

Manure Solids Content: 91.2 %

Method used to determine amount of manure:

Scale Weights

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Michael VanDerHoek (Jun 27, 2024 05:10 PDT)

Jun 27, 2024

Operator Signature | Date

Hauler Signature | Date

CALIFORNIA SOILS

№ 25226

NON CERTIFIED

lb GROSS →

lb TARE →

lb NET @ →

CARRIER: Cal Soils

TRUCK LICENSE NO: 391

TRAILER LICENSE NO: 4410

TRAILER LICENSE NO:

DRIVER: Coco

COMMODITY: Cow Manure

FROM: Van Derhoek & Sons

DEL TO: CA Soils

REMARKS:

PLEASE LEAVE WHITE AND GOLD COPIES IN THE SCALE HOUSE

CALIFORNIA SOILS

№ 25235

NON CERTIFIED

1/26/23 8:29 PM

lb GROSS → 82920 lb GR

lb TARE → 29800 lb TR

lb NET @ → 53120 lb NT
26.56 ton

CARRIER: Ca Soils

TRUCK LICENSE NO: 391

TRAILER LICENSE NO: 4410

TRAILER LICENSE NO:

DRIVER: Coco

COMMODITY: Cow Manure

FROM: Van Derhoek & Sons

DEL TO: Ca Soils

REMARKS:

PLEASE LEAVE WHITE AND GOLD COPIES IN THE SCALE HOUSE

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Michael Van Der Hoek

Name of Dairy Facility: Jelle Van Der Hoek

Facility Address:

337 Gates RD	Modesto	Stanislaus	95358
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number:

Michael Van Der Hoek	(209) 581-6133
Name	Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: California Soils

Address of Hauling Company/Person:

3401 Gaffery RD	Vernalis	CA	95385
Number and Street	City	State	Zip Code

Contact Person:

Office	(209) 835-9530
Name	Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Composting Facility

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

California Soils	(209) 835-9530
Name	Phone Number

3401 Gaffery RD	Vernalis	CA	95385
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

3401 Gaffery RD	Vernalis	95385
Address	City	Zip Code

	Stanislaus
Street and nearest cross street (if no address)	County

Assessor's Parcel Number	Assessor's Parcel Number County

Last date hauled: 03/28/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 46.40 tons

Manure Solids Content: 91.2 %

Method used to determine amount of manure:

Scale Weights

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Michael VanDerHoek (Jun 27, 2024 05:10 PDT)

Jun 27, 2024

Operator Signature | Date

Hauler Signature | Date

CALIFORNIA SOILS

№ 25153

NON CERTIFIED

lb GROSS → 76760 LB GR

lb TARE → 29800 LB TA

lb NET @ → 46960 LB NET

CARRIER Ca Soils

TRUCK LICENSE NO 391

TRAILER LICENSE NO 4410

TRAILER LICENSE NO

DRIVER Coco

COMMODITY Manure

FROM VAN OerHOOKE Sons

DEL TO Ca Soils

REMARKS

PLEASE LEAVE WHITE AND GOLD COPIES IN THE SCALE HOUSE

CALIFORNIA SOILS

№ 26759

NON CERTIFIED

lb GROSS → 76460 LB GR

lb TARE → 29700 LB TA

lb NET @ → 46760 LB NET

CARRIER Ca Soils

TRUCK LICENSE NO 395

TRAILER LICENSE NO 4410

TRAILER LICENSE NO

DRIVER Coco

COMMODITY Cow Manure

FROM Vander Hoek Dairy

DEL TO Ca Soils

REMARKS

PLEASE LEAVE WHITE AND GOLD COPIES IN THE SCALE HOUSE

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

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OPERATOR INFORMATION

Name of Operator: Michael Van Der Hoek

Name of Dairy Facility: Jelle Van Der Hoek

Facility Address:

337 Gates RD	Modesto	Stanislaus	95358
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number:

Michael Van Der Hoek	(209) 581-6133
Name	Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Avila Spreading Service

Address of Hauling Company/Person:

P.O. Box 580742	Modesto	CA	95358
Number and Street	City	State	Zip Code

Contact Person:

Lonny Avila	(209) 595-7955
Name	Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Broker

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Avila Spreading Service	(209) 595-7955
Name	Phone Number

P.O. Box 580742	Modesto	CA	95358
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

7243 Paradise RD	Modesto	95358
Address	City	Zip Code

	Stanislaus
Street and nearest cross street (if no address)	County

Assessor's Parcel Number	Assessor's Parcel Number County

Last date hauled: 05/19/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 600.00 tons

Manure Solids Content: 91.2 %

Method used to determine amount of manure:

60 Loads @ 10 Tons per load.

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Michael VanDerHoek (Jun 27, 2024 05:10 PDT)

Jun 27, 2024

Operator Signature | Date

Hauler Signature | Date

AVILA

PROFESSIONAL · RELIABLE · COURTEOUS

SPREADING SERVICE

P.O. Box 580742 Modesto, California 95358

Van Der Hoek Dairy
337 Gates Rd
Modesto, CA. 95358

Date: 5-19-23
Terms: *10 Days*
Fed ID: 77-0480211

Loads	Tons	Yards	Description	Cost
60	600	1200	Manure from dairy taken by Avila	

Total:

Past due after 10 days from date. If account unpaid within time permitted, customer agrees to pay interest at 18% per annum and attorney's fees if account is referred to an attorney or collection agency for collection.

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Michael Van Der Hoek

Name of Dairy Facility: Jelle Van Der Hoek

Facility Address:

337 Gates RD	Modesto	Stanislaus	95358
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: m Van Der Hoek (209) 581-6133
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Avila Spreading Service

Address of Hauling Company/Person:

P.O. Box 580742	Modesto	CA	95358
Number and Street	City	State	Zip Code

Contact Person: Lonny Avila (209) 595-7955
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Broker

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Avila Spreading Service (209) 595-7955
Name Phone Number

P.O. Box 580742	Modesto	CA	95358
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

7243 Paradise RD	Modesto	95358
Address	City	Zip Code

	Stanislaus
Street and nearest cross street (if no address)	County

Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 09/18/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 70.00 tons

Manure Solids Content: 52.0 %

Method used to determine amount of manure:

7 Loads @ 10 Tons per load.

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Michael VanDerHoek (Jun 27, 2024 05:10 PDT)

Jun 27, 2024

Operator Signature | Date

Hauler Signature | Date

AVILA
PROFESSIONAL · RELIABLE · COURTEOUS
SPREADING SERVICE

P.O. Box 580742 Modesto, California 95358

Van Der Hoek Dairy
337 Gates Rd
Modesto, Ca. 95358

Date: 9-18-23
Terms: *10 Days*
Fed ID: 77-0480211

Loads	Tons	Yards	Description	Cost
31	310	620	Excavated manure from lagoon stockpiled	$1,240.00
7	70	140	Manure taken to Vernalis and stockpiled	$700.00

Thank you!

Total: **$1,940.00**

Past due after 10 days from date. If account unpaid within time permitted, customer agrees to pay interest at 18% per annum and attorney's fees if account is referred to an attorney or collection agency for collection.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Van Der Hoek & Sons Dairy #2
337 Gates Rd.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 3/23/23 3:30 PM
SUBMITTED BY: Sean
ANALYZED DATE: 3/24/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 18.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW - Milk Barn	T3082017A	Electrical Conductivity (EC)	EPA 120.1 and	0.677	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	6.34	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	433	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	100			
Electrical Conductivity (EC)	101			0.470
Nitrate Nitrogen (NO3-N)	103	97.4 (T3082012A)	98.0 (T3082012A)	0.634

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Van Der Hoek & Sons Dairy #2
337 Gates Rd.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 3/23/23 3:30 PM
SUBMITTED BY: Sean
ANALYZED DATE: 3/24/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 18.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW - #2	T3082017B	Electrical Conductivity (EC)	EPA 120.1 and	0.772	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	5.56	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	494	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	100			
Electrical Conductivity (EC)	101			0.470
Nitrate Nitrogen (NO3-N)	103	97.4 (T3082012A)	98.0 (T3082012A)	0.634

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

DENELE ANALYTICAL, INC.
865 S. KILROY ROAD
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: VAN DER HOEK DAIRY
Mailing address:
(not facility address) MODESTO CA
Phone#:

Authorized Copy Released To: F&R AG SERVICES
Sampled By: Sean Locke
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
							Analysis				
1	3/23/23	11:00	DW- MILK BARN	DOMESTIC	18.0'		X				
2	"	11:05	DW- #2	"	18.0'		X				
3											
4											
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
		F&R AG SERVICES				
2nd		DENELE ANALYTICAL, INC.	3-23-23 3:3			T3082017

Required to be submitted to any agency? (YES / NO)

DELLAVALLE™
LABORATORY INC

Jelle VanDer Hoek
337 Gates Road
Modesto, CA 95358

Account# 00-0017600
Account Manager: Ben Nydam
Submitted By: F&R Ag Services

Received: 08/29/2023 15:30
Reported: 09/01/2023 16:29

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23H2333-01	IW-North	Ag Water	Sean Locke		08/29/2023 7:45

Default Cooler Temperature on Receipt °C: 3.8
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Reported: 09/01/2023 16:29

Sample Results

Sample: IW-North
23H2333-01 (Water)

Sampled: 8/29/2023 7:45
Sampled By: Sean Locke

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.68**	mmhos/cm	0.01	1		08/30/23 11:40	SM 2510 B		BEH1424
Electrical Conductivity umhos	**1680**	umhos/cm	10.0	1		08/30/23 11:40	SM 2510 B		BEH1424
Nitrate Nitrogen as NO3N	**51.6**	mg/L	0.1	1	10	08/29/23 18:56	EPA 300.0		BEH1341
pH	**7.6**	units	1.0	1		08/30/23 11:40	SM 4500-H+	H	BEH1424
Total Filterable Solids (TDS)	**1010**	mg/L	10.0	1		09/01/23 12:59	SM 2540 C		BEH1449
Temperature	**25.0**	°C	0.0	1		08/30/23 11:40	SM 2510 B		BEH1424
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/31/23 08:54	SM 4500-NH3 C		BEH1420
Total Nitrogen	**51.9**	mg/L	1.00	1		08/31/23 08:54	SM 4500-NH3 C		BEH1420

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Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1341										
Blank (BEH1341-BLK1)					Prepared & Analyzed: 8/29/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH1341-BLK2)					Prepared & Analyzed: 8/29/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEH1341-BS1)					Prepared & Analyzed: 8/29/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		103	90-110		
Duplicate (BEH1341-DUP1)		**Source: 23H2257-01**			Prepared & Analyzed: 8/29/2023					
Nitrate Nitrogen as NO3N	0.08		0.1	mg/L		0.08			1.23	10
Matrix Spike (BEH1341-MS1)		**Source: 23H2257-01**			Prepared & Analyzed: 8/29/2023					
Nitrate Nitrogen as NO3N	5.3		0.1	mg/L	5.000	0.08	105	90-110		
Reference (BEH1341-SRM1)					Prepared & Analyzed: 8/29/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEH1341-SRM2)					Prepared & Analyzed: 8/29/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		

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Account# 00-0017600
Account Manager: Ben Nydam
Submitted By: F&R Ag Services

Received: 08/29/2023 15:30
Reported: 09/01/2023 16:29

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1420										
Blank (BEH1420-BLK1)					Prepared: 8/30/2023	Analyzed: 8/31/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
Blank (BEH1420-BLK2)					Prepared: 8/30/2023	Analyzed: 8/31/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
LCS (BEH1420-BS1)					Prepared: 8/30/2023	Analyzed: 8/31/2023				
Kjeldahl Nitrogen (TKN), Total	5.63		1.00	mg/L	5.709		98.6	90-110		
LCS (BEH1420-BS2)					Prepared: 8/30/2023	Analyzed: 8/31/2023				
Kjeldahl Nitrogen (TKN), Total	5.93		1.00	mg/L	5.709		104	90-110		
Duplicate (BEH1420-DUP1)		**Source: 23H0060-01**			Prepared: 8/30/2023	Analyzed: 8/31/2023				
Kjeldahl Nitrogen (TKN), Total	14.9		3.50	mg/L		16.1			7.86	10
Duplicate (BEH1420-DUP2)		**Source: 23H2333-01**			Prepared: 8/30/2023	Analyzed: 8/31/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.40	mg/L		ND				10
Matrix Spike (BEH1420-MS1)		**Source: 23H0060-01**			Prepared: 8/30/2023	Analyzed: 8/31/2023				
Kjeldahl Nitrogen (TKN), Total	28.6		3.50	mg/L	9.990	16.1	125	90-110		
Matrix Spike (BEH1420-MS2)		**Source: 23H2333-01**			Prepared: 8/30/2023	Analyzed: 8/31/2023				
Kjeldahl Nitrogen (TKN), Total	4.29		1.40	mg/L	7.992	ND	53.7	90-110		
Reference (BEH1420-SRM1)					Prepared: 8/30/2023	Analyzed: 8/31/2023				
Kjeldahl Nitrogen (TKN), Total	22.7			mg/L	23.80		95.3	90-110		

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LABORATORY INC

Jelle VanDer Hoek
337 Gates Road
Modesto, CA 95358

Account# 00-0017600
Account Manager: Ben Nydam
Submitted By: F&R Ag Services

Received: 08/29/2023 15:30
Reported: 09/01/2023 16:29

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1424										
Blank (BEH1424-BLK1)					Prepared & Analyzed: 8/30/2023					
pH	5.5		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH1424-BLK3)					Prepared & Analyzed: 8/30/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	6.8		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Duplicate (BEH1424-DUP2)	**Source: 23H2339-02**				Prepared & Analyzed: 8/30/2023					
Electrical Conductivity	0.33		0.01	mmhos/cm		0.33			0.151	10
pH	9.1		1.0	units		9.1			0.110	10
Electrical Conductivity umhos	332		10.0	umhos/cm		332			0.151	10
Reference (BEH1424-SRM1)					Prepared & Analyzed: 8/30/2023					
Electrical Conductivity	522			umhos/cm	538.0		97.1	90-110		
Reference (BEH1424-SRM2)					Prepared & Analyzed: 8/30/2023					
pH	5.9			units	5.820		101	28178-101.7		
Reference (BEH1424-SRM3)					Prepared & Analyzed: 8/30/2023					
Electrical Conductivity	988			umhos/cm	1000		98.8	90-110		
Electrical Conductivity umhos	988			umhos/cm	1000		98.8	90-110		
Reference (BEH1424-SRM5)					Prepared & Analyzed: 8/30/2023					
Electrical Conductivity	984			umhos/cm	1000		98.4	90-110		
Electrical Conductivity umhos	984			umhos/cm	1000		98.4	90-110		
Reference (BEH1424-SRM6)					Prepared & Analyzed: 8/30/2023					
pH	4.1			units	4.000		102	97.5-102.5		
Reference (BEH1424-SRM8)					Prepared & Analyzed: 8/30/2023					
pH	4.0			units	4.000		101	97.5-102.5		

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Account# 00-0017600
Account Manager: Ben Nydam
Submitted By: F&R Ag Services

Received: 08/29/2023 15:30
Reported: 09/01/2023 16:29

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1449										
Blank (BEH1449-BLK1)					Prepared: 8/31/2023	Analyzed: 9/1/2023				
Total Filterable Solids (TDS)	ND		10.0	mg/L						
LCS (BEH1449-BS1)					Prepared: 8/31/2023	Analyzed: 9/1/2023				
Total Filterable Solids (TDS)	17.5		10.0	mg/L	2000		0.875	0-200		
Duplicate (BEH1449-DUP1)			**Source: 23H2331-01**		Prepared: 8/31/2023	Analyzed: 9/1/2023				
Total Filterable Solids (TDS)	690		10.0	mg/L		680			1.46	10
Reference (BEH1449-SRM1)					Prepared: 8/31/2023	Analyzed: 9/1/2023				
Total Filterable Solids (TDS)	323			mg/L	325.0		99.5	90-110		

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08/29/23 15:30 23H2333

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

8-29-23

JG

No. of **Water** Samples 1 No. of Bottles 12

Logged By _____ Results Needed By _____

BILL TO: 17600 / 008 Acct#/Cons. Purchase Order No

VAN DER HOEK DAIRY
Company Name

337 GATES RD.
Address

MODESTO, CA 95358
City, State Zip

Telephone Fax

Cell/Email

REQUESTED BY F&R Ag Services

PROJECT

Crop Intended

Sampled By SEAN LOCKE

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well
[ ] Supply Water

Other: No. of Samples **Soil** **Tissue** **Manure**

Analysis and Bottles Required:

[ ] Well Water **(DWW1)** - EC, NO_3-N, Field NH_4-N, if detected run NH_4-N (1-1 liter plastic, unpreserved) **(Annually)**
[ ] Well Water **(DWW2)** - DWW1 plus HCO_3, CO_3, Cl, SO_4, Dissolved: Ca, Mg, Na, TDS (1-1 liter plastic, unpreserved) **(Every 5 yrs)**
[ ] Process Water **(DPW1)** - EC, NO_3-N, NH_4-N, TKN, TDS, TP, TK (1-1 liter plastic, unpreserved) **(Quarterly)**
[ ] Process Water **(DPW2)** - DPW1 plus CO_3, HCO_3, Cl, SO_4, Total: Ca, Mg, Na, (1-1 liter plastic, unpreserved) **(Every 2 yrs)**
[X] Canal Water **(DCW1)**-EC, NO_3-N,TKN,TN,TDS (1-1 liter plastic,unpreserved)
[ ] Water – EC, NO_3-N, TDS, TP (1-1 liter plastic, unpreserved)
[ ] Manure **(DM1)** - N, P, K, Moisture **(2x/yr)**
[ ] Manure **(DM2)** - DM1 plus Ash, Ca, Mg, Na, S, Cl **(Every 2 yrs)**
[ ] Compost **(OSA2)** – Moisture, N, P, K, Ash, OM, C:N ratio, Ca, Mg, Na, Fe, Cu, Mn, Zn, S
[ ] Harvest Tissue **(CRA1)** - Moisture , N, P, K, Ash
[ ] Tissue – % Moisture
[ ] Fall Soil – **(DS2)** - FA1 (0-1') **plus CEC, OM**; NO_3-N (1-2')
[ ] OTHER:

COPIES TO: labs@fragservices.com, slocke@fragservices.com
[ ] Co Health Dept
[ ] RWQCB [X] Copy of Chain
[ ] State Forms [ ] QA/QC Documents

Description of Sample	Field NH_4-N	Date/Time Sampled	Rec'd Temp °C
1. IW- NORTH		8/29 7:45	3.8
2.			
3.			
4.			
5			
6.			
7.			
8.			
9.			
10.			

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 9/26/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	[signature]	F+R	8/29 7:45	
Second	JLF	DLV	8-29-23 2:00	8-29-23 3:08
Third				
Fourth	[signature]	DVL	8/29 1530	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24%) or $5 00 per month whichever is greater

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc , it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc (*cal*) If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory

Invoicing Information:

Shipping
$_____ In
$_____ Out

Amt Paid Rec By Check # Date

Signature

08/29/23 15:30 23H2333

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

☐ Samples refridgerated before pick up | ☐ Picked up samples placed in Ice chest

Container: Ice Chest ☒ Box ☐ None ☐ | **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☒ Preserved Upon Receipt at Laboratory

	Type of Container(s) Received	Sample Number 1	2	3	4	5	6	7	8	9	10
	Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
Plastics	250 mL unpreserved (White) Plastic	1									
Plastics	250 mL HNO_3 (Red) Plastic										
Plastics	* pH Value										
Plastics	250 mL H_2SO_4 (Yellow) Plastic	1									
Plastics	* pH Value	<2									
Plastics	500 mL unpreserved (White) Plastic										
Plastics	1 L unpreserved (White) Plastic										
Plastics	1 L unpreserved (BOD) (Purple) Plastic										
Special	500mL unpreserved (White) Glass										
Special	PO4-P Kit										
Special	**Other:**										
	Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
Plastics	250 mL unpreserved (White) Plastic										
Plastics	250 mL HNO_3 (Red) Plastic										
Plastics	250 mL H_2SO_4 (Yellow) Plastic										
Plastics	500 mL HNO_3 (Red)										
Plastics	1 L unpreserved (White) Plastic										
Plastics	1 L unpreserved (BOD) (Purple) Plastic										
Plastics	1 L HNO_3 (Red)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ + MCAA (EPA531)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ (EPA547)										
VOA Vials	40mL AG VOA unpreserved (White) (Set of 3)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
VOA Vials	40mL VOA, H_3PO_4 (Set of 3)										
VOA Vials	40 mL VOA, HCl (Blue) (Set of 3)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
Glass	250 mL AG H_2SO_4 (Yellow)										
Glass	250 mL AG $Na_2S_2O_3$ (Green)										
Glass	250 mL AG $Na_2S_2O_3$ + MCAA										
Glass	500 mL glass unpreserved (White)										
Glass	500 mL AG HCl (Blue)										
Glass	1 L AG unpreserved (White)										
Glass	1 L AG H_2SO_4 (Yellow)										
Glass	1 L AG $Na_2S_2O_3$ (Green)										
Glass	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
Special	Cyanide - 500 mL NaOH										
Special	Asbestos - 1L P wrapped in foil (Set of 2)										
Special	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
Special	Chlorite/Bromate - 250 mL AG with EDA										
Special	HAA5 - 250mL AG Ammonium Chlorite										
Special	DO KIT										
Special	**Other:**										
Special	**Other:**										

pH Strips
Lot: 10BDH4501 Exp: Jan 2025

SEND OUT

2023 Van der Hoek Annual Report

Final Audit Report 2024-06-27

Created:	2024-06-22
By:	Sean Locke (slocke@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAA-2Xaep8xfdAJjtduwHYdohNXnlVMTAUz

"2023 Van der Hoek Annual Report" History

Document created by Sean Locke (slocke@fragservices.com)
2024-06-22 - 3:15:07 PM GMT

Document emailed to mkdairy@verizon.net for signature
2024-06-22 - 3:17:08 PM GMT

Email viewed by mkdairy@verizon.net
2024-06-27 - 12:04:25 PM GMT

Signer mkdairy@verizon.net entered name at signing as Michael VanDerHoek
2024-06-27 - 12:10:25 PM GMT

Document e-signed by Michael VanDerHoek (mkdairy@verizon.net)
Signature Date: 2024-06-27 - 12:10:27 PM GMT - Time Source: server

Agreement completed.
2024-06-27 - 12:10:27 PM GMT

Adobe Acrobat Sign